Transformations, Sums, and Mixtures

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Module 2: Generate

Computational Methods in Statistics and Data Science (DATASCI 406)

Transformations

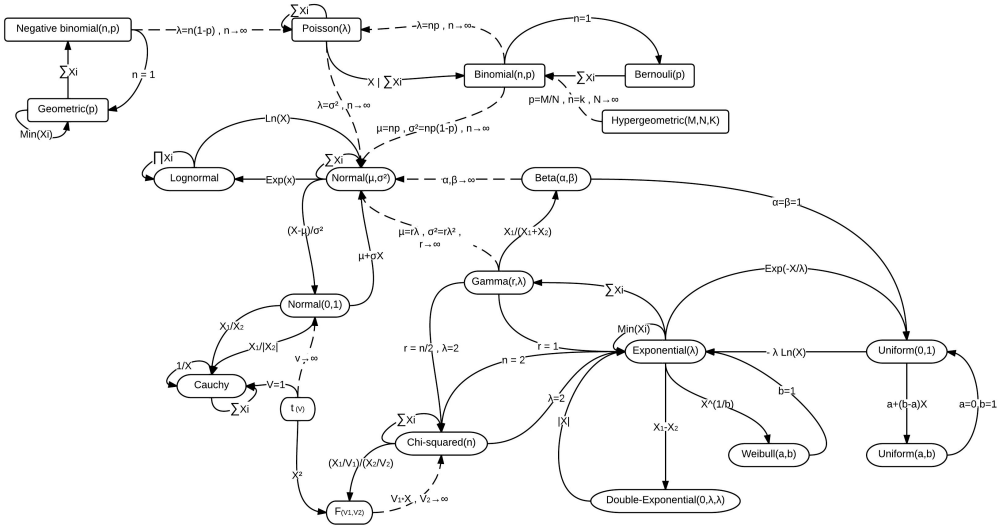
A new variable

Suppose we have some random variable X with a very complex $Q(u)$ so that the inversion method is difficult to implement.

For example, the Normal distribution's $F(x)$ cannot be written in closed form, so we can't easily invert it.

Then we must search for **transformations** of random variables that we can sample from.

General strategy: **show** $F(x) = G(x)$, where G is the CDF of the transformation.



Source: Wikipedia, "Relationships among probability distributions."

Creating variables from functions of other variables

Sometimes we define **RVs as functions of other random variables**.

Examples:

- The "log Normal" distribution:

$$X = \exp(Y), \quad Y \sim N(\mu, \sigma^2)$$

- The χ^2_k distribution: $\sum_{i=1}^{k} Z_i^2, Z_i \overset{\text{iid}}{\sim} N(0,1)$
- The t_k distribution: $Z/\sqrt{V/k}$, with $Z \sim N(0,1)$ independent of $V \sim \chi^2_k$
- Test statistics/estimators: $f(X_1, X_2, \ldots, X_n)$

More difficult case: Working backwards

Suppose we want to sample X, can we find transformations that lead to it?

Tips and tricks:

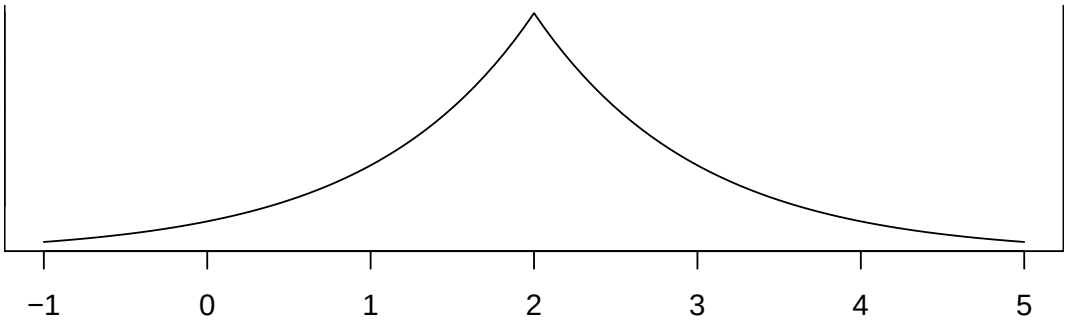
- Reduce to a simpler case
- Write down $f(x)$ or $F(x)$ and find patterns.
- Try conditioning on some event, then randomly generate that event.
- Calculus techniques on $f(x)$ (change of variables)

Laplace Distribution

Here is the PDF of the **Laplace distribution**:

$$f(x) = \frac{1}{2} \exp\{-|x - \theta|\}$$

For example, when $\theta = 2$



Example: Laplace distribution

To generate:

$$X \sim \text{Laplace}(\theta)$$

Claim: X can be generated by a transformation:

$$X = SM + \theta$$

where:

$$P(S = 1) = 0.5, P(S = -1) = 0.5, \quad M \sim \text{Exp}(1)$$

Where does this come from?

Reducing to a simpler case

First, let's deal with the **parameter** θ.

Let $Y \sim \text{Laplace}(\theta)$ with CDF:

$$P(Y \leq t) = \int_{-\infty}^{t} \frac{1}{2} e^{-|y-\theta|}\, dy$$

Perform a **change of variables**, $x = y - \theta$ so $y = x + \theta$ and $dy = dx$, so that

$$P(Y \leq t) = \int_{-\infty}^{t-\theta} \frac{1}{2} e^{-|x|}\, dx = P(X \leq t - \theta) = P(X + \theta \leq t)$$

where $X \sim \text{Laplace}(0)$. In other words, Y is a **Laplace(0) added to** θ.

Using symmetry of X

Notice that X (with $\theta = 0$) is **symmetric about 0**:

$$f(x) = \frac{1}{2}\exp\{-|x|\} = f(-x) \Rightarrow P(X < 0) = P(X > 0) = \frac{1}{2}$$

We can use the **disjoint events** $X < 0$ and $X > 0$ to write:

$$P(X \leq t) = P(X \leq t \mid X < 0)P(X < 0) + P(X \leq t \mid X > 0)P(X > 0)$$

By the definition of conditional probability:

$$P(X \leq t \mid X > 0) = \frac{P(X \leq t, X > 0)}{P(X > 0)} = \frac{P(0 < X \leq t)}{P(X > 0)}$$

Notice that (for $t > 0$):

$$P(X \leq t \mid X > 0) = \frac{\int_0^t f(x)\,dx}{P(X > 0)} = \int_0^t \frac{(1/2)\exp\{-|x|\}}{1/2}\,dx = \int_0^t \exp\{-x\}\,dx$$

Recognizing CDF

We saw that $X \mid X > 0$ had the CDF:

$$P(X \leq t \mid X > 0) = \int_0^t \exp\{-|x|\}\, dx = \int_0^t \exp\{-x\}\, dx$$

We recognize that $X \mid X > 0 = M \sim \text{Exp}(1)$ and by symmetry we see that $X \mid X < 0$ is $-M$.

Let A be the event that $X > 0$, which has $P(A) = 1/2$.

- When A occurs $X = M$.
- When A^c occurs, $X = -M$
- In other words, if $P(S = 1) = P(S = -1) = 1/2$, then $X = SM$

Putting it all together

We first saw that if

$$Y \sim \text{Laplace}(\theta) \Rightarrow Y = X + \theta, X \sim \text{Laplace}(0)$$

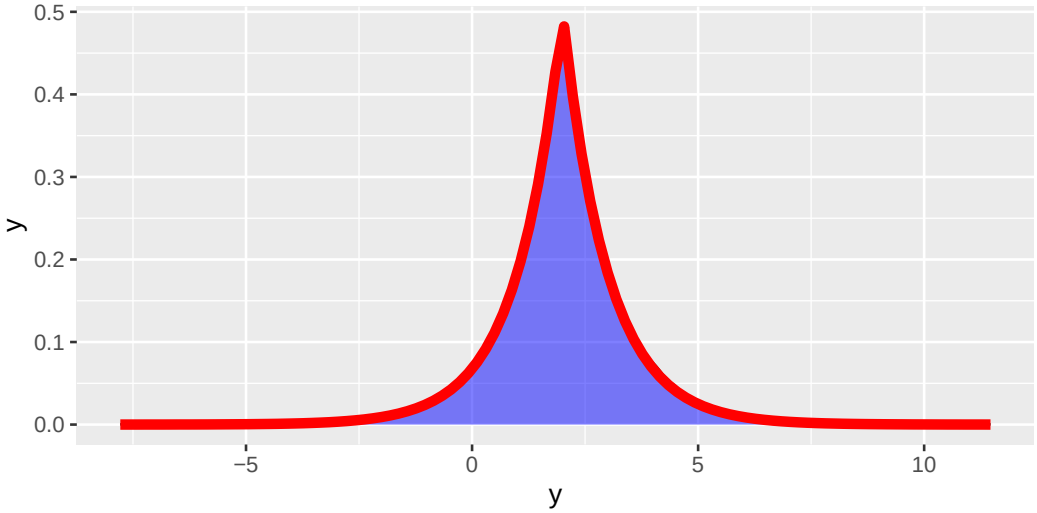
How to generate Laplace(0)? We found that could be represented as:

$$X = SM, M \sim \text{Exp}(1), P(S = 1) = P(S = -1) = 1/2$$

In R:

```
rlaplace <- function(n, theta) {
    sample(c(-1, 1), n, replace = TRUE) * rexp(n) + theta }
```

Checking results



Transformations Summary

We want to sample from X, but can't (or don't want to) use the quantile method. If we know $f(x)$ or $F(x)$, we can sometimes work backwards using these techniques:

- Reduce to a simpler case
- Write down $f(x)$ or $F(x)$ and find patterns.
- Try conditioning on some event, then randomly generate that event.
- Calculus techniques on $f(x)$ (change of variables)

No single algorithm will always work, so this method requires a lot of creativity.

Sums and Convolutions

Linear combinations of variables

Two ways of combining variables linearly are particularly common:

- Linear combination of random variables (convolutions): $X = \sum a_i Y_i$
- Linear combination of distribution functions (mixtures): $F_X(t) = \sum a_i F_i(t)$ where $0 \leq a_i \leq 1$ and $\sum a_i = 1$.

They are **not equivalent**. With independent $Y_1 \sim N(-2, 1)$ and $Y_2 \sim N(2, 1)$, the average $(Y_1 + Y_2)/2 \sim N(0, 1/2)$ has one mode, while the 50/50 mixture of the two has two.

We cover sums first, then mixtures.

Convolutions

For independent X and Y with densities, $Z = X + Y$ has the density

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(z - y) f_Y(y)\, dy,$$

which is called the **convolution** of f_X and f_Y.

The integral is often hard to compute, but **sampling Z is easy**: draw X and Y and add them.

If a target distribution can be written as a sum of variables we can already draw, we have a sampler for it.

Gamma as a sum of exponentials

If $E_1, \ldots, E_k \overset{\text{iid}}{\sim} \text{Exp}(\lambda)$, then

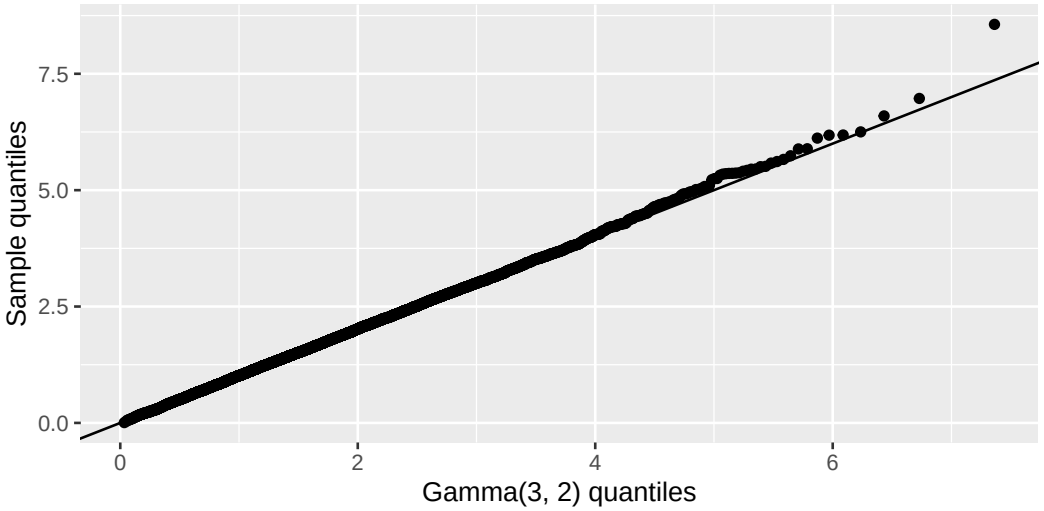
$$\sum_{i=1}^{k} E_i \sim \text{Gamma}(k, \lambda).$$

With the exponential sampler from video 1, $E_i = -\log(U_i)/\lambda$:

```
rgamma_sum <- function(n, k, rate) {
  e <- matrix(-log(runif(n * k)) / rate, nrow = k) # one column per draw
  colSums(e)
}
```

This works only for **integer** shape k.

Checking the Gamma sampler



Binomial as a sum of Bernoullis

If $B_1, \ldots, B_m \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$, then $\sum_{i=1}^{m} B_i \sim \text{Bin}(m, p)$. A Bernoulli draw is $U \leq p$.

```
rbinom_sum <- function(n, size, prob) {
  b <- matrix(runif(n * size) <= prob, nrow = size)
  colSums(b)
}
x <- rbinom_sum(10000, size = 5, prob = 0.3)
round(rbind(sampled = table(factor(x, levels = 0:5)) / 10000,
            exact = dbinom(0:5, 5, 0.3)), 3)
```

```
##             0     1     2     3     4     5
## sampled 0.168 0.355 0.306 0.137 0.031 0.003
## exact   0.168 0.360 0.309 0.132 0.028 0.002
```

Each draw costs `size` uniforms, so for large `size` inversion is cheaper.

Mixture Distributions

Defining a mixture

A **mixture** of distributions $F_1, \ldots, F_k$ has the CDF

$$F(x) = \sum_{i=1}^{k} a_i F_i(x), \qquad 0 \le a_i \le 1, \quad \sum_{i=1}^{k} a_i = 1.$$

The F_i are the **components** and the a_i are the **mixing weights**.

The same distribution can be perceived in two ways:

- **Latent variable**: an unobserved C with $P(C = i) = a_i$ picks the component. Draw C, then draw X from F_C.
- **Random parameter**: the components are one family $F(x; \theta)$, and θ is itself random. Draw θ, then draw X from $F(x; \theta)$.

Basic properties of mixtures

Proof that $F(t) = \sum_{i=1}^{k} a_i F_i(t), 0 \leq a_i \leq 1, \sum_{i=1}^{k} a_i = 1$ is a valid distribution

- Bounded above by 1 (and likewise below by 0):

$$\lim_{t\to\infty} F(t) = \lim_{t\to\infty} \sum_{i=1}^{k} a_i F_i(t) = \sum_{i=1}^{k} a_i \lim_{t\to\infty} F_i(t) = \sum_{i=1}^{k} a_i = 1$$

- Nondecreasing: for $t_2 > t_1$,

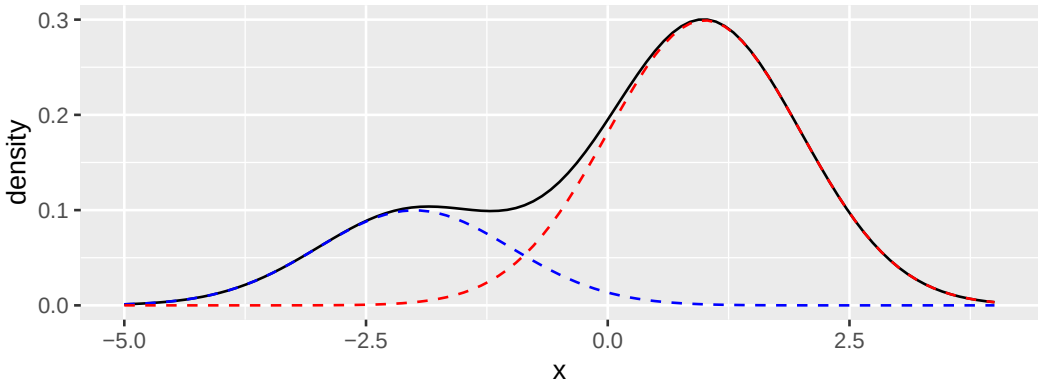
$$a_i F_i(t_1) \leq a_i F_i(t_2) \Rightarrow \sum_{i=1}^{k} a_i F_i(t_1) \leq \sum_{i=1}^{k} a_i F_i(t_2) \Rightarrow F(t_1) \leq F(t_2)$$

- If all F_i have density f_i,

$$\frac{d}{dx} F(x) = \sum_{i=1}^{k} a_i \frac{d}{dx} F_i(x) = \sum_{i=1}^{k} a_i f_i(x) = f(x)$$

Example: Bimodal Normal

$$f(x) = 0.25\phi(x+2) + 0.75\phi(x-1)$$



The dashed curves are the weighted components $0.25\phi(x+2)$ and $0.75\phi(x-1)$.

Mixtures: “latent variables”

A **latent variable** is a random variable that **is not observed.** If $P(C = i) = a_i$ and $X \mid C = i \sim F_i$, then

$$P(X \leq t) = \sum_{i=1}^{k} P(X \leq t \mid C = i) P(C = i) = \sum_{i=1}^{k} a_i F_i(t).$$

For the bimodal Normal, C picks the mode:

```
k <- 10000
C <- rbinom(k, size = 1, prob = 0.75)
mix_latent <- ifelse(C == 1, rnorm(k, mean = 1), rnorm(k, mean = -2))
```

The Laplace distribution is also a latent-variable mixture, of Exp(1) and $-$Exp(1).

Mixtures: “random parameters”

When the components are one family $F(x;\theta)$, we draw θ first and then $X \mid \theta \sim F(x;\theta)$.

The bimodal Normal is $N(\mu, 1)$ where μ is 1 with probability 0.75 and -2 otherwise:

```
mu <- sample(c(1, -2), k, replace = TRUE, prob = c(0.75, 0.25))
mix_param <- rnorm(k, mean = mu)
```

The parameter can also vary **smoothly**, $\theta \sim G$, which gives

$$F(x) = \int F(x;\theta)\, g(\theta)\, d\theta.$$

For example, $\theta \sim U(0,1)$ and $X \mid \theta \sim \text{Bin}(10, \theta)$.

Example: Noisy Data

Suppose you are sampling data that you believe is (iid)

$$Y_i \sim N(\mu, 1)$$

Your goal is to test a hypothesis about μ, for example:

$$H_0 : \mu = 0 \quad \text{vs} \quad H_1 : \mu \neq 0$$

Suppose about 1 in 5 of our data gets contaminated by adding

$$Y_i + \epsilon_i, \quad \text{where } \epsilon_i \sim N(0, 9)(\text{iid})$$

Our observed data is either $N(\mu, 1)$ (with probability 0.8) or $N(\mu, 10)$ (with probability 0.2).

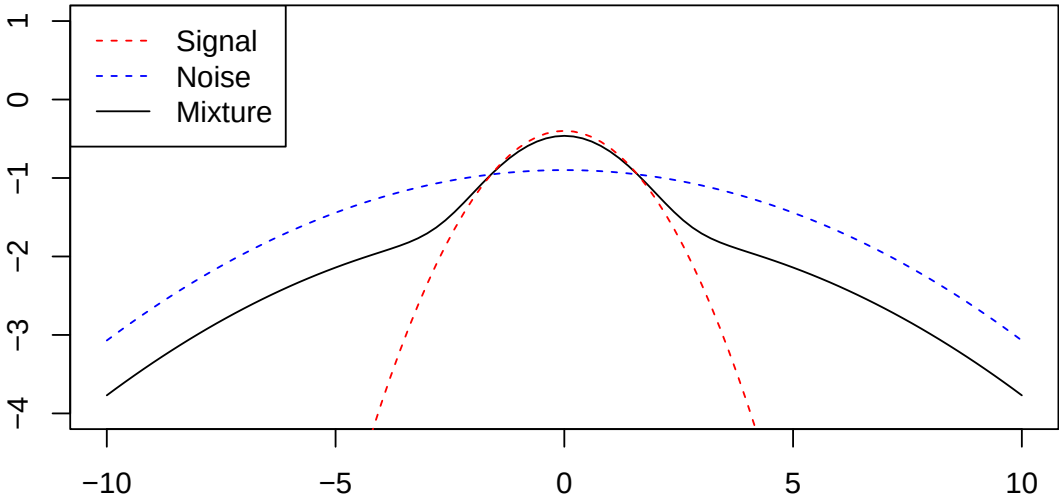
Contaminated data is a mixture

With a latent $B_i \sim \text{Bernoulli}(0.2)$ marking the contaminated observations,

$$X_i = Y_i + B_i \epsilon_i, \qquad F_X(x) = 0.8F(x; \mu, 1) + 0.2F(x; \mu, 10).$$

```
rnoisy <- function(n, mu) {
    y <- rnorm(n, mean = mu, sd = 1)
    epsilon <- rnorm(n, mean = 0, sd = 3)
    b <- rbinom(n, size = 1, prob = 0.2)
    y + b * epsilon
}
```

Mixture compared to the components (log scale)



Testing without contamination

We want to test the following hypothesis at the $\alpha = 0.05$ level:

$$H_0 : \mu = 0 \quad \text{vs} \quad H_1 : \mu \neq 0$$

Suppose **we didn't know about the contamination** so that we thought:

$$X_i \sim N(0, 1) \quad \text{when } H_0 \text{ is true}$$

If we use the **sample mean** as the **test statistic**, we would think that the following **rejection rule** would have proper α-level:

$$\text{Reject if } \left|\bar{X}\right| \geq 1.96/\sqrt{n} \text{ and accept otherwise.}$$

Does our rejection rule have proper size?

Observe that even with contamination, the null hypothesis $H_0 : \mu = 0$ would hold if $X_i \sim (0.8)N(0,1) + (0.2)N(0,10)$. Do we reject this null hypothesis too often? Not enough?

Procedure for testing the Type I error rate:

- Pick α-level ($\alpha = 0.05$)
- Find a rejection region ($|\bar{X}| \geq 1.96/\sqrt{n}$)
- Sample from the null hypothesis. (e.g., 1000 groups of 10 each)
- Compute the test statistic distribution.
- How often is the hypothesis rejected?

Group the data generated under the null that $H_0 : \mu = 0$ into 1000 groups of 10:

```
noisy_groups <- replicate(1000, rnoisy(10, 0), simplify = FALSE)
```

Compute the test statistic distribution:

```
noisy_means <- map_dbl(noisy_groups, mean) # distribution of T
```

Compute the proportion of test statistics that lead to rejecting the true null hypothesis:

```
(noisy_size <- mean(abs(noisy_means) >= qnorm(0.975, sd = 1/sqrt(10))))
```

```
## [1] 0.212
```

We reject the true null about $4\times$ too often!

Achieving proper size

In a sense, this wasn't really a fair fight. We **ignored the contamination**.

Let's find an actual α-level rejection region (instead of using the one from $N(0, 1)$).

```
(noisy_mean_cutoffs <- quantile(noisy_means, c(0.025, 0.975)))
```

```
##   2.5%  97.5%
## -1.115  1.070
```

Let's validate that this has proper size:

```
mean(noisy_means <= noisy_mean_cutoffs[1] |
     noisy_means >= noisy_mean_cutoffs[2])
```

```
## [1] 0.05
```

Trimmed Means

We know that about 20% of our data are contaminated and the contamination is equally likely to be positive or negative.

Idea: what if we dropped the smallest 10% and the largest 10% of our data before computing the mean?

We've seen the **trimmed mean** before.

```
trimmed_mean <- function(x, p) {
  xqs <- quantile(x, c(p/2, 1 - (p/2)))
  keep(x, x > xqs[1] & x < xqs[2]) %>% mean
}
```

Distribution of the trimmed mean for noisy data

```
noisy_trimmed <- map_dbl(noisy_groups, trimmed_mean, p = 0.2)
(noisy_trimmed_cutoffs <- quantile(noisy_trimmed, c(0.025, 0.975)))
```

```
##    2.5%   97.5%
## -0.8548  0.8796
```

Power when $\mu = 1$

Let's compare the power of the test using the

- standard sample mean
- 20% trimmed mean

when $H_1 : \mu = 1$ is actually true and the α-level is 0.05.

Procedure:

- Find the **rejection region corresponding to the null**.
- Generate data **according to the alternative hypothesis**.
- Compute the proportion of time **the test statistic is in the rejection region**.

Rejection regions:

```
noisy_mean_cutoffs
```

```
##   2.5%  97.5%
## -1.115  1.070
```

```
noisy_trimmed_cutoffs
```

```
##    2.5%   97.5%
## -0.8548  0.8796
```

Distributions when the alternative is true (i.e., $\mu = 1$):

```
noisy_alt_data <- replicate(1000, rnoisy(10, mu = 1), simplify = FALSE)
noisy_alt_means <- map_dbl(noisy_alt_data, mean)
noisy_alt_trim  <- map_dbl(noisy_alt_data, trimmed_mean, p = 0.2)
```

Proportion of statistics that fall in the cutoff region:

```
mean(noisy_alt_means <= noisy_mean_cutoffs[1] |
     noisy_alt_means >= noisy_mean_cutoffs[2])
```

```
## [1] 0.456
```

```
mean(noisy_alt_trim <= noisy_trimmed_cutoffs[1] |
     noisy_alt_trim >= noisy_trimmed_cutoffs[2])
```

```
## [1] 0.616
```

Trimmed mean is more powerful!

Linear combination of CDFs:

$$F(x) = \sum_{i=1}^{k} a_i F_i(x), 0 \leq a_i \leq 1, \sum_{i=1}^{k} a_i = 1$$

or

$$F(x) = \int_{-\infty}^{\infty} F(x;\theta) g(\theta)\, d\theta \quad \theta \sim G$$

- Easy to reason about **CDFs and PDFs** (but quantile functions not easy, even if $Q_i(u)$ is known for all i).
- Generate using **latent variables**, **random parameters**.
- Useful for common way random processes might combine.